Advanced Portfolio Management – Evaluation

You have been assigned with the task of designing a long-term portfolio strategy for a private wealth management client based in the US with assets worth USD 100 million. The portfolio must meet the following conditions:

- It should be invested in different asset classes (Bonds, Commodities, Equities, Money Market, Precious Metals, Real Estate), with a minimum investment of 20% in bonds or money market instruments (no constraint on other asset classes)
- No short sales
- The portfolio should consist of ETFs investing in different asset classes, with a minimum history of 10 years
- Choose ETFs that are representative of their asset class, do not attempt to choose the "best" ETF in each class. Keep the number of ETFs low.
- You must show the frontier to your client so they understand the risk-return trade-off
- Find also the portfolio with the maximum expected return subject to a 1-month maximum value at risk constraint of 3% with a 99% confidence on the portfolio return (you may assume normality or use the historical VaR)
- You must at least discuss estimation risk. Implementing Black-Litterman or any other correction method is optional.

Tip: find ETFs investing in different asset classes here: https://www.justetf.com/en/etf-lists.html, specifically in the "List of ETFs by asset class." Clicking on an asset class, you will access the screener. You may add a column for inception date to make sure you select ETFs at least 10 years old. Use their tickers with yahoo finance to download the data and work with monthly returns.

Prepare a report of not more than 5 pages with your proposal. Justify your choices. In particular, justify the choice of ETFs and asset classes in terms of their role in diversification and expected performance. Do not submit your code.

Have fun!